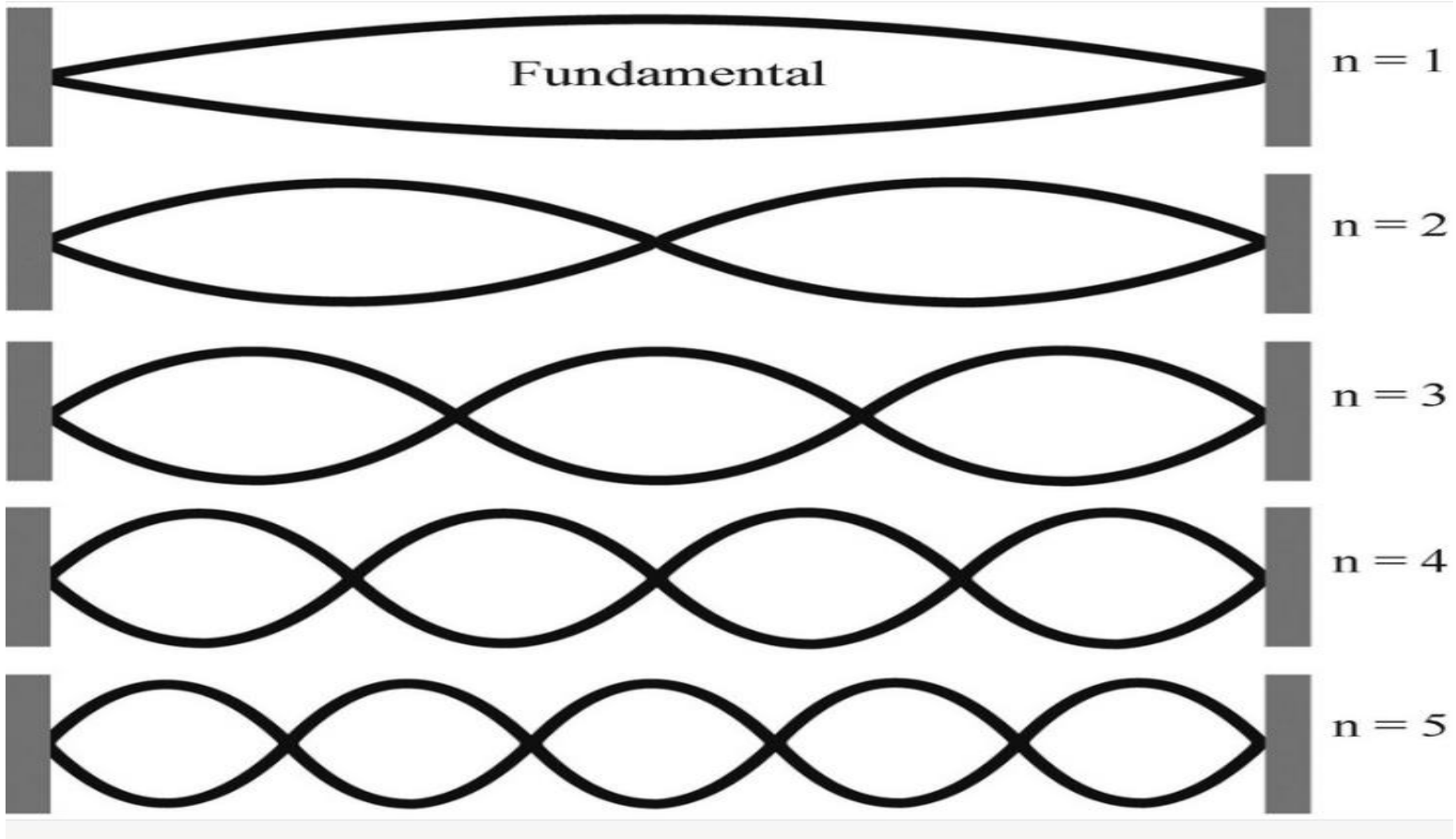
Lista de exercícios - Física Geral 2- Ondas

- A lista de exercícios vale 30% da segunda avaliação parcial.

- O prazo de entrega é 29/09/2022 sala de aula.

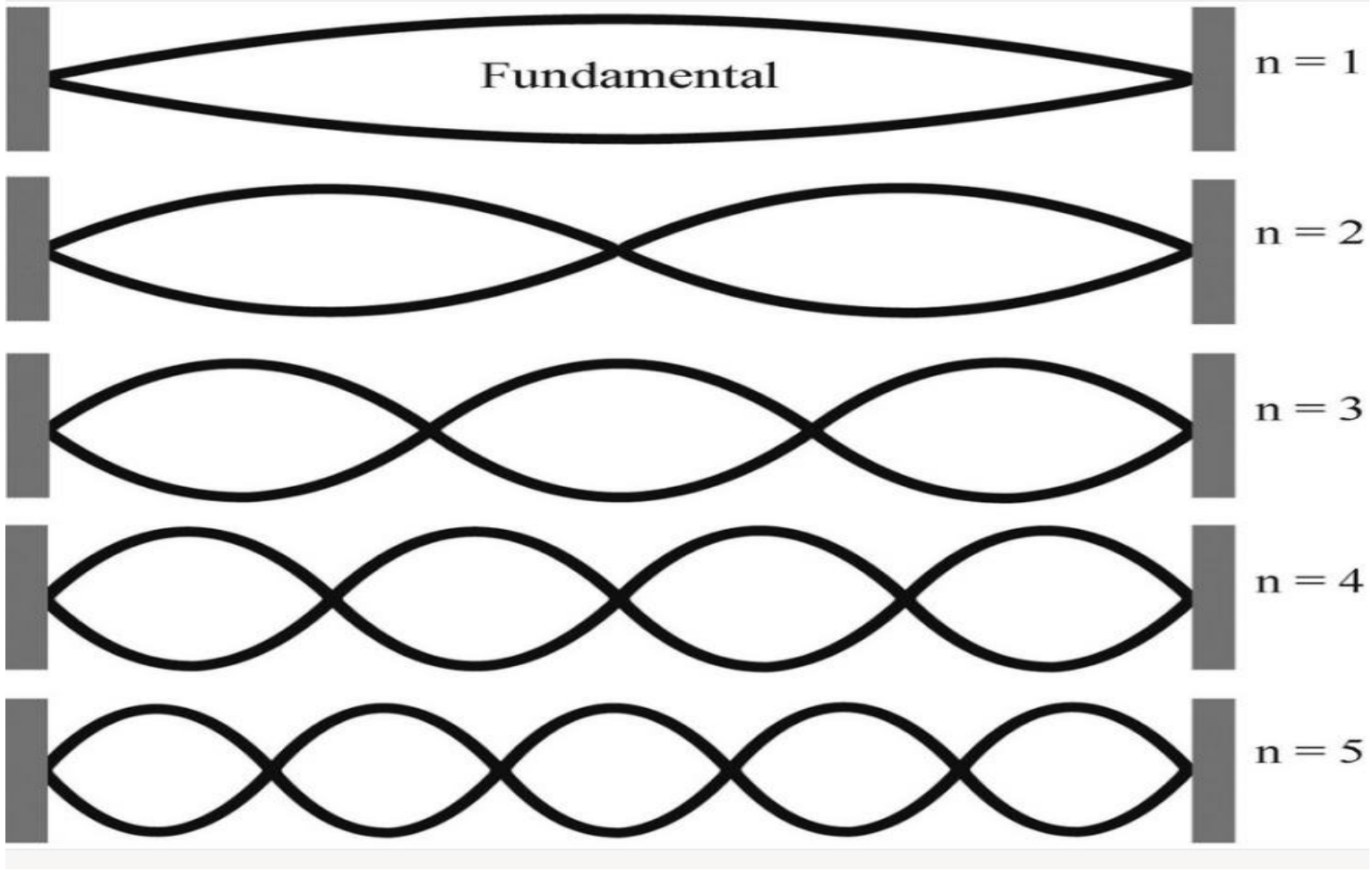
- (1) Dada a equação de onda unidimensional com condições de fronteira:

$$\frac{\partial^2 y(x,t)}{\partial t^2} - v^2 \frac{\partial^2 y(x,t)}{\partial x^2} = 0. \quad \longrightarrow \quad y(0,t) = y(L,t) = 0$$



- Mostramos que a função de onda y(x,t) é dada por

$$y(x,t) = \sum_{n=1}^{\infty} \left(D_n^+ e^{i\frac{n\pi}{L}vt} + D_n^- e^{-i\frac{n\pi}{L}vt} \right) \sin \frac{n\pi x}{L}.$$



Baseados nas condições iniciais

$$y(x,t) = \sum_{n=1}^{\infty} \left(D_n^+ e^{i\frac{n\pi}{L}vt} + D_n^- e^{-i\frac{n\pi}{L}vt} \right) \sin \frac{n\pi x}{L}.$$

$$\boxed{\begin{aligned} y(x,0) &= F(x), \\ \left.\frac{\partial y(x,t)}{\partial t}\right|_{t=0} &= V(x). \end{aligned}}$$

$$\begin{aligned} F(x) &= \sum_{n=1}^{\infty} (D_n^+ + D_n^-) \sin \frac{n\pi x}{L}, \\ V(x) &= \sum_{n=1}^{\infty} i\frac{n\pi}{L} v (D_n^+ - D_n^-) \sin \frac{n\pi x}{L}. \end{aligned}$$

- (1) Mostre que os coeficientes D_n^+ e D_n^- satisfazem

$$\begin{cases} D_n^+ + D_n^- &= \dfrac{2}{L}\displaystyle\int_0^L F(x)\sin\frac{n\pi x}{L}dx \\ D_n^+ - D_n^- &= -\dfrac{2i}{n\pi v}\displaystyle\int_0^L V(x)\sin\frac{n\pi x}{L}dx, \end{cases}$$

$$\begin{aligned} D_n^+ &= \frac{1}{L}\int_0^L F(x)\sin\frac{n\pi x}{L}dx - \frac{i}{n\pi v}\int_0^L V(x)\sin\frac{n\pi x}{L}dx \\ &= \frac{1}{L}F_n - \frac{i}{n\pi v}V_n \end{aligned}$$

$$\begin{aligned} D_n^- &= \frac{1}{L}\int_0^L F(x)\sin\frac{n\pi x}{L}dx + \frac{i}{n\pi v}\int_0^L V(x)\sin\frac{n\pi x}{L}dx \\ &= \frac{1}{L}F_n + \frac{i}{n\pi v}V_n, \end{aligned}$$

(2) Baseado na função função de onda:

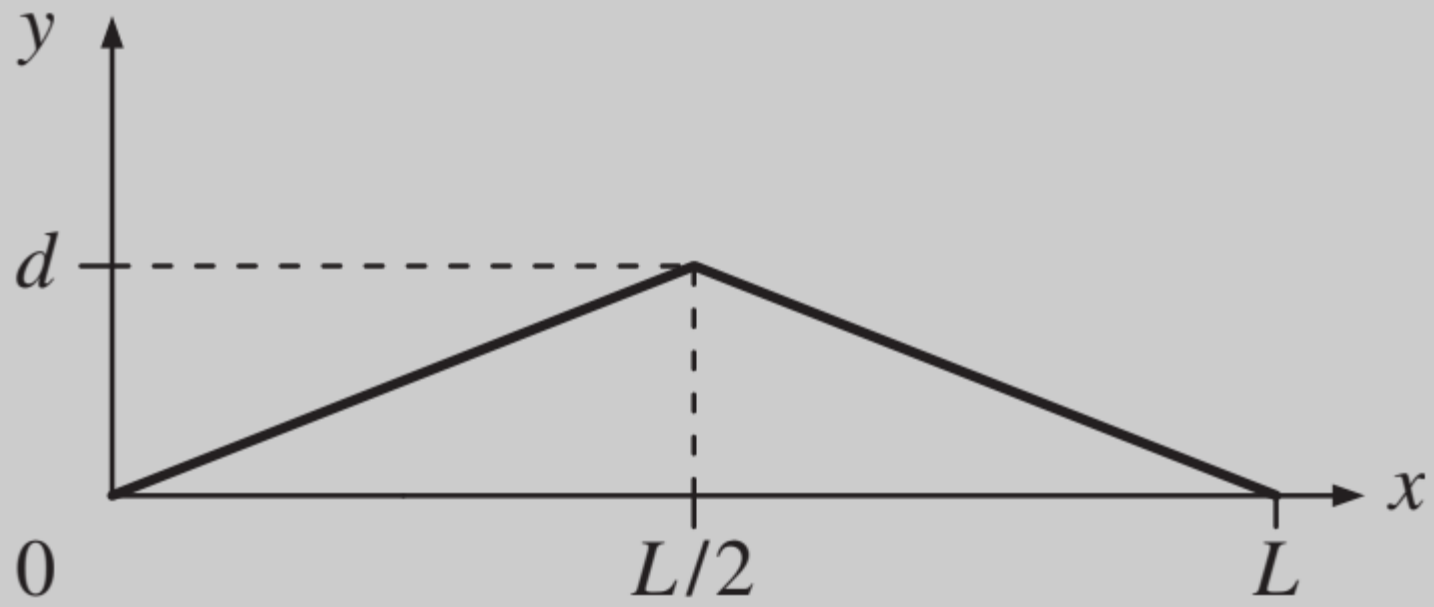
$$y(x,t) = \frac{2}{L}\sum_{n=1}^{\infty} F_n \cos\frac{n\pi v t}{L}\sin\frac{n\pi x}{L},$$

$$F_n = \int_0^L F(x)\sin\frac{n\pi x}{L}dx$$

Determine as amplitudes $A_n = \frac{2}{L}F_n$

se a condição inicial é

$$F(x) = f(x) = \begin{cases} f(x) = \dfrac{2d}{L}x, & 0 \le x \le L/2, \\ f(x) = 2d - \dfrac{2d}{L}x & L/2 \le x \le L. \end{cases}$$



$$A_n = \frac{2}{L}\left[\int_0^{L/2} \mathrm{d}x\, f(x) \sin\left(\frac{n\pi}{L}x\right) + \int_{L/2}^{L} \mathrm{d}x\, f(x) \sin\left(\frac{n\pi}{L}x\right)\right].$$

$$\frac{2}{L}\left[\int_0^{L/2} \mathrm{d}x \left(\frac{2d}{L}x\right) \sin\left(\frac{n\pi}{L}x\right) + \int_{L/2}^{L} \mathrm{d}x \left(2d - \frac{2d}{L}x\right) \sin\left(\frac{n\pi}{L}x\right)\right]$$

$$= \frac{2}{L}\left[\frac{2d}{L}\int_0^{L/2} \mathrm{d}x\, x \sin\left(\frac{n\pi}{L}x\right) + 2d\int_{L/2}^{L} \mathrm{d}x \sin\left(\frac{n\pi}{L}x\right)\right.$$

$$\left. - \frac{2d}{L}\int_{L/2}^{L} \mathrm{d}x\, x \sin\left(\frac{n\pi}{L}x\right)\right].$$

Dica: Use os seguintes resultados para as integrais, depois avalie os limites dessas integrais

$$\int \mathrm{d}x \sin ax = -\frac{1}{a}\cos ax, \quad \int \mathrm{d}x\, x \sin ax = \frac{1}{a^2}\sin ax - \frac{x}{a}\cos ax,$$

$$A_n = \frac{8d}{(n\pi)^2}\sin\left(\frac{n\pi}{2}\right).$$

$$n = 1, \qquad A_1 = \frac{8d}{(\pi)^2}$$

$$n = 3, \qquad A_3 = -\frac{8d}{(3\pi)^2}$$

$$n = 5, \qquad A_5 = \frac{8d}{(5\pi)^2}$$

(3) Two very long strings are tied together at the point $x = 0$. In the region $x < 0$, the wave speed is v_1, while in the region $x > 0$, the speed is v_2. A sinusoidal wave is incident from the left ($x < 0$); part of the wave is reflected and part is transmitted. For $x < 0$, the displacement of the wave is describable by $y(x,t) = A \sin(k_1 x - \omega t) + B \sin(k_1 x + \omega t)$, while for $x > 0$, $y(x,t) = C \sin(k_2 x - \omega t)$, where $\omega/k_1 = v_1$ and $\omega/k_2 = v_2$. (*a*) If we assume that both the wave function y and its first spatial derivative $\partial y/\partial x$ must be continuous at $x = 0$, show that $C/A = 2/(1 + v_1/v_2)$, and that $B/A = (1 - v_1/v_2)/(1 + v_1/v_2)$. (*b*) Show that $B^2 + (v_1/v_2)C^2 = A^2$.